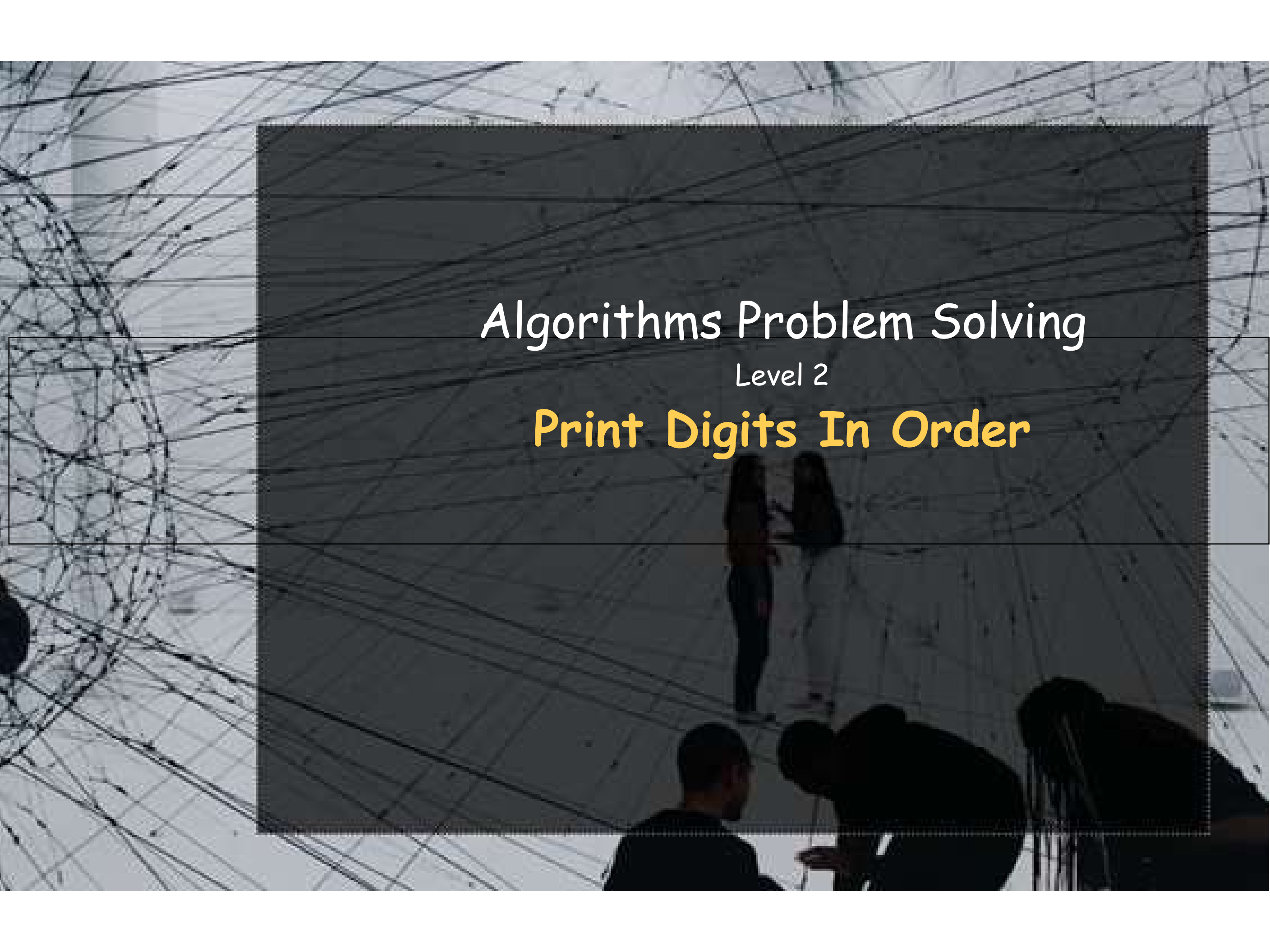


Algorithms Problem Solving

Level 2

Print Digits In Order

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower part of the image, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 2

Print Digits In Order

Definition

- ◆ Printing digits in order means displaying the digits of a number from left to right, starting with the first digit and ending with the last digit.
- ◆ For example, `1234` is printed as `1`, `2`, `3`, `4`.

Explanation

- ◆ ``ReadPositiveNumber()`` reads the number as a ``string`` instead of an ``int``.
- ◆ Reading the number as a string allows the program to access each digit directly from left to right.
- ◆ The ``do-while`` loop makes sure the entered value is not negative and is not ``0``.
- ◆ ``number[0]`` accesses the first character of the string.
- ◆ ``number[0] == '-'`` checks whether the number starts with a negative sign.
- ◆ ``number == "0"`` prevents the user from entering zero.
- ◆ ``PrintDigits()`` uses a range-based ``for`` loop to go through every character in the string.
- ◆ Each character is printed using ``cout << digit << endl``.

Steps in main()

1. Read the number as a string.
2. Check that the number is positive.
3. Pass the string to `PrintDigits()`.
4. Use a range-based `for` loop to access each character.
5. Print each digit from left to right.

💡 Example

◆ Input:

``1234``

◆ The string contains:

``'1' → '2' → '3' → '4'``

◆ The range-based ``for`` loop processes them in the same order.

➡ Result

``1``

``2``

``3``

``4``

Note

- ◆ Unlike the previous lessons, this program does not use ``% 10`` or ``/ 10`` to extract digits.
- ◆ Because the number is stored as a string, each digit can be accessed directly from left to right.
- ◆ ``char digit`` represents one character from the string.

⊘ Warning

- ◆ The validation only checks whether the first character is `'-'` or whether the entire input is `"0"`.
- ◆ It does not fully validate that every character entered is a digit. For example, an input such as `12a4` would pass the current validation.



Comparison

▨ Using `string`

- ✓ Digits can be processed directly from left to right.
- ✓ No `% 10` or `/ 10` is needed.
- ✓ The original order is preserved.

▤ Using `int`

- ✓ Digits can be processed using `% 10` and `/ 10`.
- ✓ These operations naturally extract digits from right to left.
- ✓ Additional logic would be needed to print them from left to right.

✦ Summary

- ✓ Read the number as a `string`.
- ✓ Validate that the number is positive.
- ✓ Use a range-based `for` loop to access each character.
- ✓ Print each digit in its original order.
- ✓ No `% 10` or `/ 10` is required.

Thank you 😊